Ground state of the Hubbard model with a spin-dependent external potential

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Abstract

We investigate the competition between attractive spin-spin interactions and spin-separating external forces in the ground state of a one-dimensional Fermi-Hubbard model. We consider a lattice with open boundary conditions, subject to a linear external potential whose gradient β is opposite for the two spin components. We find that as β is increased, the bound pairs start breaking one-by-one, and three distinct regimes can be identified (pairs not broken, pairs partially broken, all pairs broken), separated by critical threshold gradients. We also verify that this pair-breaking process persists under harmonic confinement. Our results are testable in cold-atom experiments.

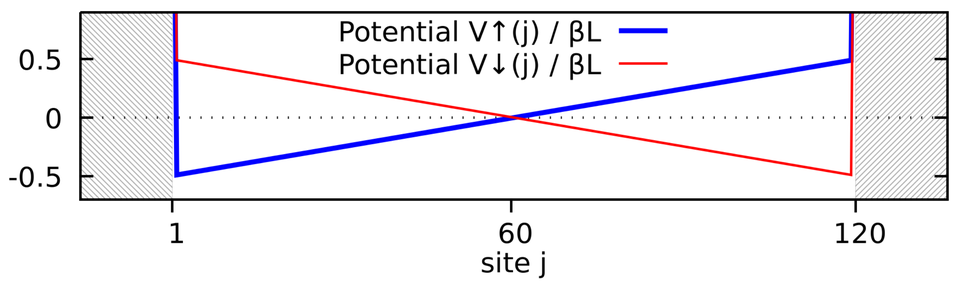
The model

The setup: A one-dimensional lattice with L sites, with open boundary conditions (*i.e.* hard-wall). Fixed fermion numbers $N\uparrow = N\downarrow = N/2$. Attractive on-site interactions $U < 0$

External potential: Linear potential with gradient β, with opposite sign for the two spin components

$$\hat{H} = -\sum_{j=1}^{L}\left(c^{\dagger}_{j,\uparrow}c_{j+1,\uparrow} + c^{\dagger}_{j,\downarrow}c_{j+1,\downarrow} + \mathrm{H.c.}\right) + U\sum_{j=1}^{L} n_{j,\uparrow}n_{j,\downarrow}$$
$$+ \beta\sum_{j=1}^{L}\left(j - \frac{L+1}{2}\right)(n_{j,\uparrow} - n_{j,\downarrow})$$

External potential for $\beta > 0$:



Finding the ground state: We use the Density Matrix Renormalization Group (DMRG) numerical method

Results of tuning the potential

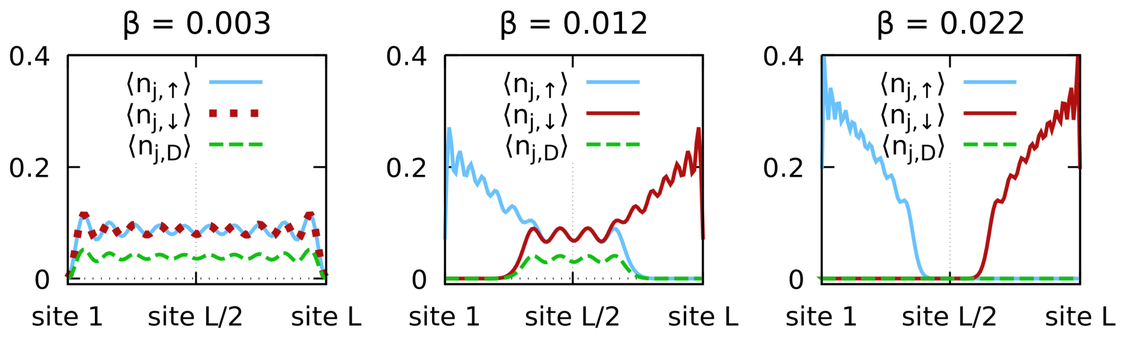
Ground state for uniform system (β=0): The Bethe ansatz solution says: all particles join up into $N/2$ bound pairs of opposite-spin fermions

Ground state for stronger gradient (β>0): As we increase β, the bound pairs break, one by one — their constituent fermions move to the lattice edges. The one-body densities, and the total doublon number, change in abrupt jumps.

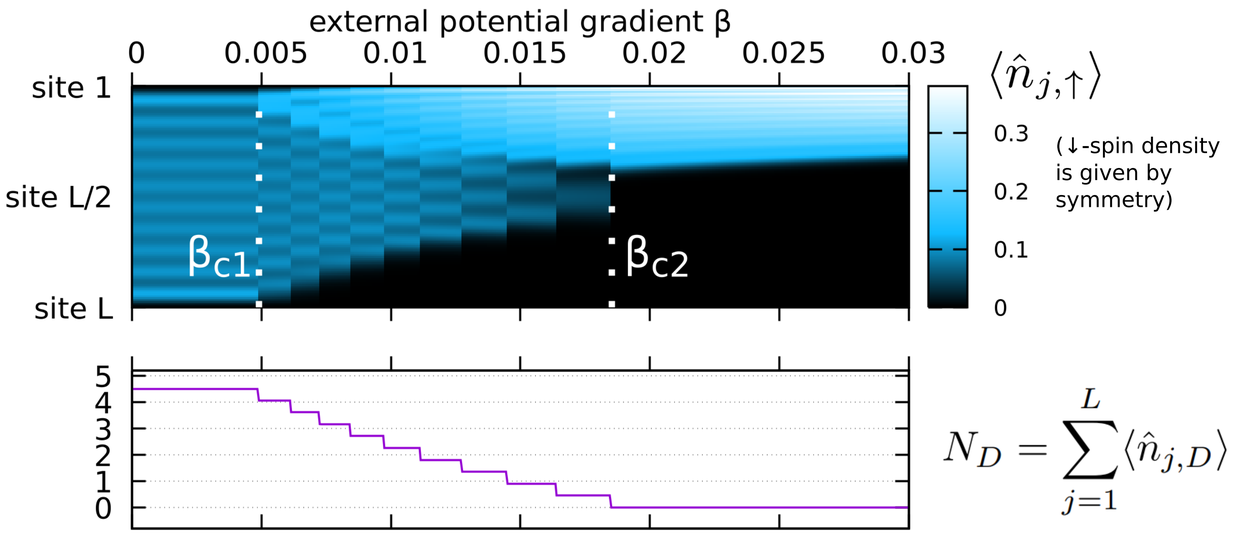
Critical values of β: βc1 — first pair breaks; βc2 — last pair breaks. Therefore, three distinct regimes of β can be established.

Example results for L = 120, U = –2, N = 10+10

For three example values of β, the densities of up-spin fermions ($n_{j,\uparrow}$), down-spin fermions ($n_{j,\downarrow}$), and doubly-occupied sites ($n_{j,D}$) are as follows:



For β tuned in a wide range, the ↑-spin density and the total doublon number change as follows:



$$N_D = \sum_{j=1}^{L}\langle \hat{n}_{j,D}\rangle$$

Counting the number of bound pairs

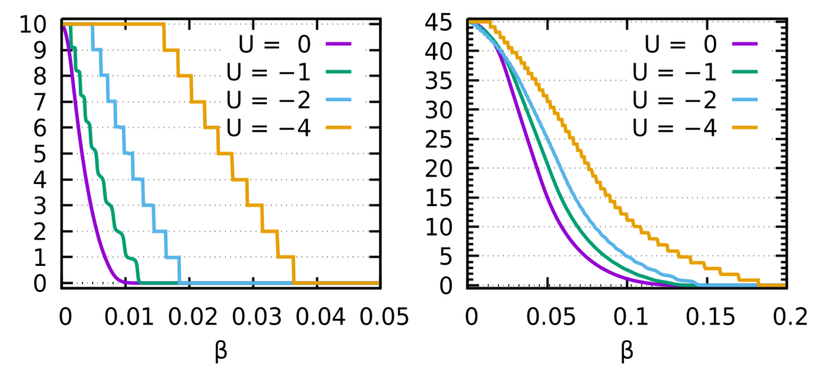
Doublon number and bound pairs: There is <1 doublon per bound pair. This ratio stays approximately unchanged, even as β is increased.

Counting the bound pairs: By rescaling the doublon number — so that it is equal to $N/2$ for β=0 — we can track the number of bound pairs as β is tuned.

The estimated pair number drops in integer plateaus.

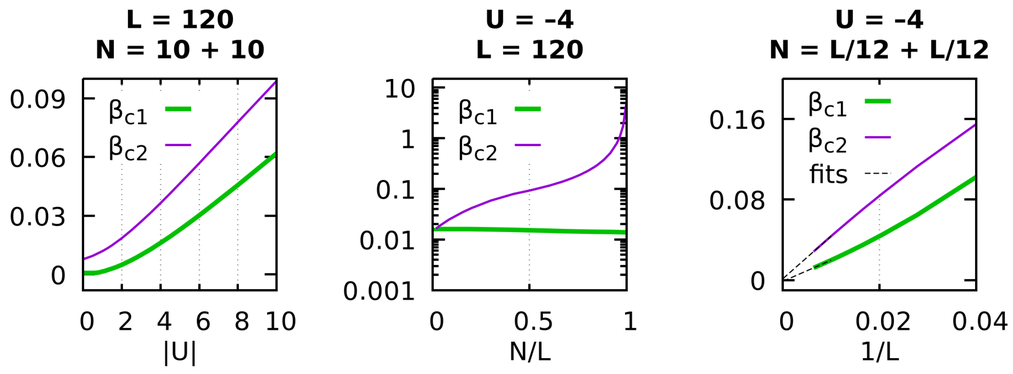
(At small $|U|$, the pairs are extended over several sites, and they become looser with increasing β. The steps in N_D are smoothed out)

N_D rescaled:



Critical values of the gradient β

What determines the thresholds βc1, βc2: The critical values of β depend directly on the system parameters. It's even possible to derive approximate formulas that connect βc1, βc2 to U, L, N (see our preprint for details).



Harmonic confinement

Result of adding a harmonic trap: We add a spin-independent harmonic potential term with strength v:

$$v \times \left(j - \frac{L+1}{2}\right)^2$$

We still observe one-by-one pair-breaking:

N_D rescaled:

Summary

- In this Hubbard model, tuning the potential gradient allows to precisely change the total number of bound pairs.
- There are direct relationships between the system parameters and the critical gradient threshold values.
- These results can be observed also in a lattice with harmonic confinement (easier to obtain in experiments than a box-like confinement).

More details — *arXiv:2604.24068*